CS3081–ArtificialIntelligence • Spring2026

Assignment 2

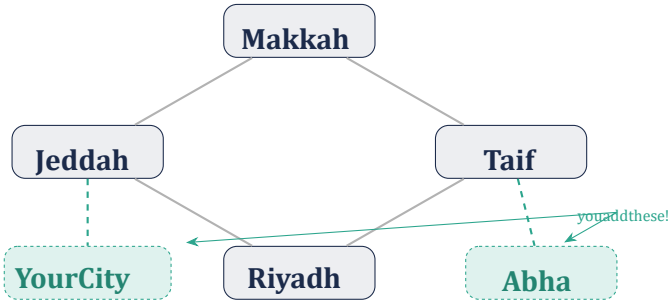
AI in My City

Instructor: **Dr. Naila Marir**

Type: **Individual – Home Assignment**

Estimated time: **1–2 Hours**

Files given: assign2 part1 search.py assign2 part2 _knowledge.py logic.py



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1. Overview

WhatisthisAssignment?

Thisassignmentextendsexactlywhatyoudidin **Lab3** .

- **Part1:** Youwillextendthecitymapbyadding2newcities,thenrunBFSto findaroute.
- **Part2:** YouwillcompleteaKnowledgeBasebyadding1missingsymboland2 missingclues.

Allfilesarealreadymostlywritten.Youonlyneedtocompletethe **TODO** lines.

FilesYouNeed

1. Copythethreefilesintoyour `Lab2` folder(samefolderasthelab).
2. OpenVSCode → openyour `Lab2` folder.
3. Openeachfileandlookfortheorange `# TODO` comments.
4. **DonotchangeanylinethatdoesnothaveaTODOcomment.**

File	What to do	TODOs
assign2 part1 _search.py	Add 2 cities, set your goal city	4
assign2 part2 _knowledge.py	Add 1 symbol + 2 clues to the KB	4
logic.py	**Do NOT touch this file**	0

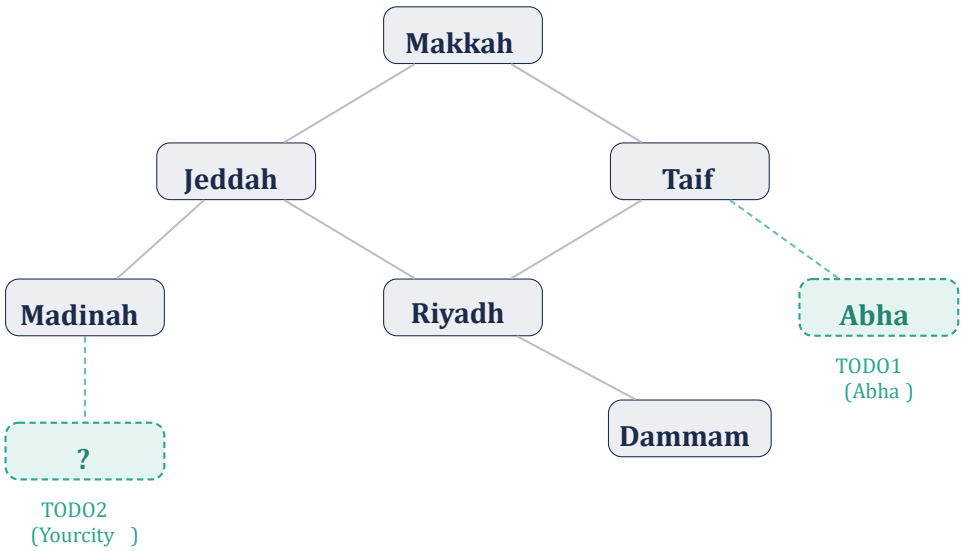
2. Part 1 – Search: Extend the City Map

2.1. Your Task

Open assign[1] part1 search.py.

The map already has 6 cities: Makkah, Jeddah, Taif, Riyadh, Madinah, Dammam. You will add **2 new cities** and search for a route to one of them.

[1] **.2. The Four TODOs**



TODO1–AddAbha(line ≈ 24)

Abhaisconnectedto **Taifonly** .

Writeonelineinthe `city map` dictionary:

```
"Abha": "Taif"],
```

Thenscrollupandadd `"Abha"` toTaif'sneighborlisttoo.

TODO2–Addyourowncity(line ≈ 31)

PickanySaudicitythatis **notalreadyinthemap** . Connectittoatleastoneexistingcity.

Example(ifyouchooseYanbu):

```
"Yanbu": "Madinah"],
```

Alsoaddyourcitytotheneighborlistofthecityitconnectsto.

TODO3–start cityisalreadyset

`start_city = "Makkah"` isalreadywritten. **Donotchangeit.**

TODO4–Setgoal city(line ≈ 79)

Set `goal_city` tothenameof **yourowncity** fromTODO2.

Example:

```
goal_city = "Yanbu"
```

2.3. Run the File

After completing all 4 TODOs, open the terminal in VS Code and run:

```
python assign2_part1_search.py
```

You should see BFS printing the cities it visits, then the final path.

2.4. Part 1 Answer Sheet

Fill in your answers below **after** running the program.

Answer1–YourNewCity

ThenameofthecityIadded(TODO2):

Cityname: Najran

Itisconnectedto: Abha

Answer2–BFSResult

Fillintheblanksusingwhatyousawintheterminaloutput.

a. BFSstartedat: Makkah

b. BFSwaslookingfor: Najran

c. ThepathBFSfoundwas:

Makkah → Taif → Abha → Najran

(addmorearrowsifyourpathislonger)

d. Thepathhad 3 steps(hops)intotal.

e. DidBFSvisitanycitiesthatwereNOTonthefinalpath? **Yes** ☐ ✓ **No** ☐

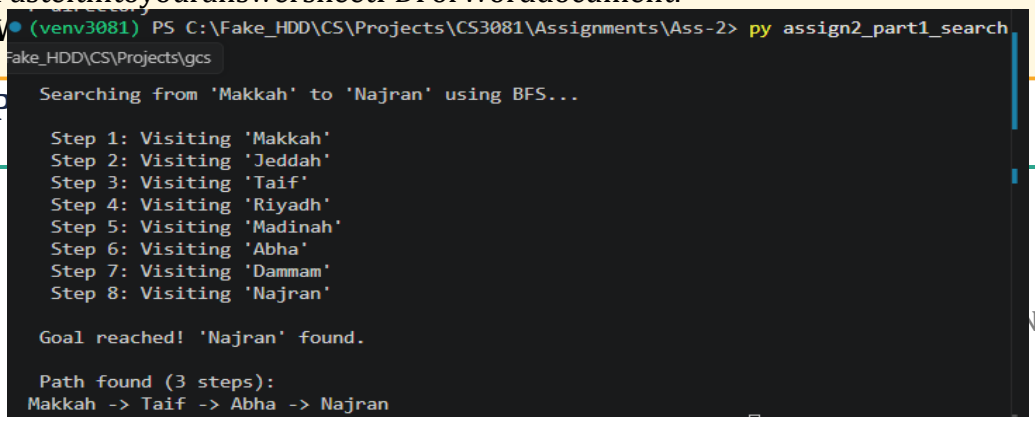
Ifyes,nameone: Jeddah

f. WhydoesBFSvisitextracitiesthatarenotonthefinalpath?

It visits all cities on the same level before going deeper in one path

AttachYourScreenshot

TakeascreenshotoftheterminaloutputafterrunningPart1.
PasteitintoyouranswersheetPDFForWorddocument.



3. P

3.1. Your Task

Open assign2 part2 knowledge.py.

The Knowledge Base already has 4 clues and 5 symbols. You need to:

- Add **1 missing symbol** (nora_suspicious)
- Add **2 missing clues** (Clues 5 and 6) to the KB

3.2. The Story

The Case: A laptop went missing from the CS lab at Effat University.

Three suspects: **Sara**, **Lina**, **Nora**

Known clues:

Clue 1: If Lina has a key → Lina is guilty. (already in KB) **Clue 2:** Sara does *not* have a key. (already in KB) **Clue 3:** Lina was seen near the lab. (already in KB)

Clue 4: If Lina was seen → Lina has a key. (already in KB) **Clue 5: Nora has an alibi.** (you add this – TODO 3)

Clue 6: If Nora does NOT have an alibi → Nora is suspicious. (you add this – TODO 4)

3.3. The Four TODOs

TODO 1 – Define the missing symbol (line ≈ 37)

A symbol for **"Nora is suspicious"** is missing. Add it:

```
nora_suspicious = Symbol("NoraIsSuspicious")
```

Copy this line exactly. Place it after the other symbols.

TODO 2 – Add the symbol name to all symbols (line ≈ 44)

The list `all_symbols` must include every symbol name.
Add `"NoraIsSuspicious"` to the list:

```
all_symbols = [
    "SaraHasKey",
    "LinaHasKey",
    "LinaSeenNearLab",
    "LinaIsGuilty",
    "NoraHasAlibi",
    "NoraIsSuspicious",     # <-- add this line
]
```

TODO 3 – Add Clue 5 to the KB (line ≈58)

Clue 5 says: *"Nora has an alibi."* This is a direct fact – just tell the KB the symbol.

Hint: In the lab, we added `lina_seen` as a direct fact like this:

```
kb.tell(lina_seen)
```

Do the same for `nora alibi`.

TODO 4 – Add Clue 6 to the KB (line ≈64)

Clue 6 says: *"If Nora does NOT have an alibi → Nora is suspicious."*

Use `Implication` and `Not`:

```
kb.tell(Implication(Not(nora_alibi), nora_suspicious))
```

Copy this line exactly.

3.4. Run the File

After completing all 4 TODOs, run:

```
python assign2_part2_knowledge.py
```

You should see 3 query answers printed.

3.5. Part2AnswerSheet

Answer3–QueryResults

Writetheansweryourprogramprintedforeachquery.

a. Query1–IsLinaguilty?
Programanswered: **YES** ☑ **NO** ☐

b. Query2–IsNorasuspicious?
Programanswered: **YES** ☐ **NO** ☑

c. Query3–DoesSarahaveakey?
Programanswered: **YES** ☐ **NO** ☑

Answer4–UnderstandingtheResults

a. WhyisLinaguiltyaccordingtotheKB?(Tracetheclues:whichclues leadtothisconclusion?)
Lina is seen near the lab (clue 3) which implies that she has a key (clue 4) which implies that she is guilty (clue 1).

b. Clue5saysNoraHASanalibi.YetQuery2asksifsheissuspicious. Basedontheprogram'sanswer–isshesuspicious?
No, she is not suspicious.

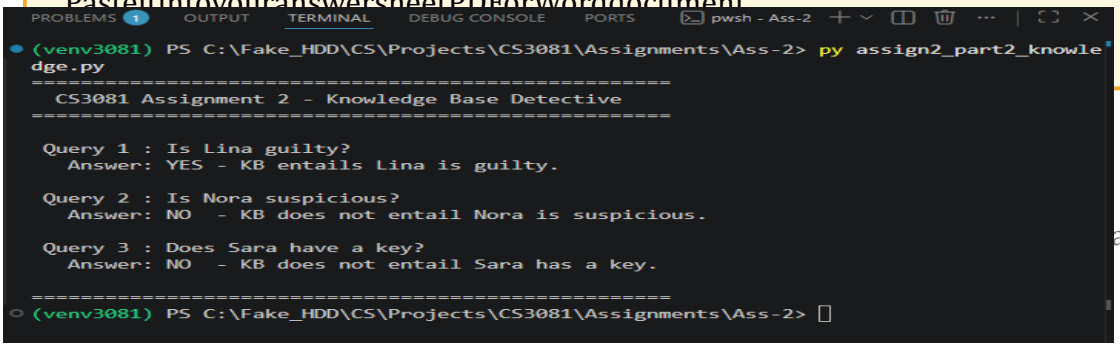
Inyourownwords,why?
Nora has an alibi, so clue 6 doesnt imply she is suspicious.

c. Completethissentenceinyourownwords:

"TheKnowledgeBasecannotproveastatementunless..."
the knowledge base entails the statement

AttachYourScreenshot

TakeascreenshotoftheterminaloutputafterrunningPart2.
PasteitintoyouranswersheetPDForWorddocument.



4. Reflection Question

This question connects **both parts** of the assignment. Answer in 2–4 sentences.

Reflection–Searchvs.Knowledge

InPart1,BFS searched forapath.InPart2,theKB deduced afact.

WhatisthedifferencebetweenwhatBFSdoesandwhattheKBdoes?
Thinkabout:Whatistheinput?Whatistheoutput?Whatkindofproblemdoes eachsolve?

They are only alike in that they are both methods of artificial intelligence where a computer thinks and behaves the same way a human brain would. BFS is a search algorithm used for things like navigation and puzzles to find paths. KB is a collection of facts that a query can be run against to deduce facts. They each have their own input, outputs, and uses.

5. Submission & Grading

5.1. What to Submit

Submit **one ZIP file** named A3_YourName_ID.zip containing:

Item	Details
assign2_part1_search.py	Your completed file with all 4 TODOs filled in
assign2_part2_knowledge.py	Your completed file with all 4 TODOs filled in
Answer Sheet (PDF or Word)	All answers + screenshots for both parts + reflection

5.2. Grading Rubric

Item	Marks	Notes
Part 1 – Code correct (TODOs 1–4)	20	Program runs without errors
Part 1 – Answer 1 (new city)	10	City name + connection
Part 1 – Answer 2 (BFS result)	15	Path + explanation
Part 2 – Code correct (TODOs 1–4)	20	Program runs, 3 queries shown
Part 2 – Answer 3 (query results)	10	Matches program output
Part 2 – Answer 4 (understanding)	15	Correct reasoning
Reflection question	10	2–4 sentences, connects both parts
Total	**100**	

Common Mistakes to Avoid

- **Forgetting to update both sides of the connection.** If you add "Abha": ["Taif"], you must also add "Abha" inside Taif's list. Otherwise BFS cannot travel back.
- **Wrong quotes or spelling.** Python is case-sensitive. "NoraIsSuspicious" and "noraisSuspicious" are different.

- **Missing the screenshot.** A correct answer without a screenshot loses all marks for that part.
- **Changing lines that are not TODOs.** Only edit lines with `# TODO`.

Good luck! You already know how to do this.

Dr. Naila Marir

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